

CS2012E Operating Systems Assignment 1

Implementation of a Character Device Driver in Linux

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Date	08/04/2026

Problem Statement

Title: Implementation of a Character Device Driver in Linux

The objective of this assignment is to design and implement a character device driver in Linux that performs controlled user operations within a specified time limit.

The driver accepts two input module parameters:

- **Kernel_version:** an array parameter specifying the kernel version
- **Time:** the time in seconds before which the specified tasks must be completed

The driver module will only be inserted if the given kernel version matches the version used during compilation. It is verified using **LINUX_VERSION_CODE**.

If insertion was successful, the driver carries out the following functions

- Register the character device and display its major and minor numbers along with timer value.
- Allow the user to interact with the device through standard file operations read and write.
- Ensure the device is first read from the userspace before writing the username of the current logged in user on to the device.
- Capture username written by user during the write operation.
- Verify whether both operations were completed successfully within the specified time limit.

Upon removal of the device when '**rmmod**' is called, the driver should evaluate the sequence and timing of operations and display either a success message with the provided username or a failure message.

Methodology

Implementation of device driver was done as follows:

1. Module Parameter Handling

The driver accepts inputs using

1. **module_param_array** for the kernel version
2. **module_param** for timer

The parameters are passed to the driver during insertion using the '**insmod**' command.

2. Validating Kernel Version

The kernel version is extracted using **LINUX_VERSION_CODE** from which the major and minor version numbers are extracted.

These values are compared with the linux kernel version used during compilation. The device is loaded only if the versions match.

3. Device Registration

The character device is registered with the help of:

- **alloc_chrdev_region()** to allocate major and minor numbers

- **cdev_init()** to initialize the device structure
- **cdev_add()** to add the device to the kernel

The assigned major and minor numbers are printed using **'printk'**.

4. Tracking Time Limit

Record the system time using **jiffies** at the time of module insertion.

Then calculate the elapsed time upon removal of the module by comparing the stored start time and the current value of **jiffies**.

5. Read Operation

The **my_read()** function becomes active when the user reads the device file using commands like **cat**, **read**, etc.

Upon calling the read function:

- The **read_flag** is set to 1 indicating the read operation has been completed.
- The event is logged using **printk**.

6. Write Operation

The **my_write()** function becomes active when the user writes to the device using commands like **echo**, **write**, etc.

The write function:

- Checks if the read operation has already been performed
- Uses **copy_from_user()** to safely transfer data from user space to kernel space
- Stores the username provided
- Sets **write_flag** to 1 indicating completion of the write operation

7. Verification of Necessary Conditions on Removal of Device

During module removal using **rmmmod**, the following conditions are verified

- If read operation was completed by checking **read_flag**
- If write operation was completed by checking **write_flag**
- Whether the operations occurred in the correct order
- Whether the tasks were completed within the specified time limit

8. Printing Output

- A success message is printed along with the provided username if all conditions are satisfied
- Otherwise, a failure message is displayed

All outputs are logged using **printk** and viewed using the **dmesg** command.

9. Cleaning Up the Device

The device is removed using:

- **cdev_del()**
- **unregister_chrdev_region()**

This ensures no resources are left allocated after removing the module.

Test Cases

Test Case 01: Successful execution (Success)

1. Running make

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ make
make -C /lib/modules/6.11.4-301.fc41.x86_64/build M=/home/kdevika/Documents/OS_S4/Assignment_1 modules
make[1]: Entering directory '/usr/src/kernels/6.11.4-301.fc41.x86_64'
warning: the compiler differs from the one used to build the kernel
The kernel was built by: gcc (GCC) 14.2.1 20240912 (Red Hat 14.2.1-3)
You are using:          gcc (GCC) 14.3.1 20250523 (Red Hat 14.3.1-1)
CC [M] /home/kdevika/Documents/OS_S4/Assignment_1/mymodule.o
MODPOST /home/kdevika/Documents/OS_S4/Assignment_1/Module.symvers
CC [M] /home/kdevika/Documents/OS_S4/Assignment_1/mymodule.mod.o
LD [M] /home/kdevika/Documents/OS_S4/Assignment_1/mymodule.ko
BTF [M] /home/kdevika/Documents/OS_S4/Assignment_1/mymodule.ko
Skipping BTF generation for /home/kdevika/Documents/OS_S4/Assignment_1/mymodule.ko due to unavailability of vmlinux
make[1]: Leaving directory '/usr/src/kernels/6.11.4-301.fc41.x86_64'
```

2. Insert the module

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo insmod mymodule.ko kernel_version=6,11 time_limit=60
[sudo] password for kdevika:
```

3. Check kernel logs for device registration

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo dmesg
[ 8913.172635] Device registered, Major=235, Minor=0, Time=60
```

4. Create device file

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo mknod /dev/mychardev c 235 0
```

5. Perform read

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ cat /dev/mychardev
```

6. Perform write

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ echo "devika" | sudo tee /dev/mychardev
devika
```

7. Remove module

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo rmmod mymodule
```

8. Verify final output

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo dmesg
[ 8913.172635] Device registered, Major=235, Minor=0, Time=60
[ 8924.395043] Device read completed successfully
[ 8928.321583] Username recieved: devika
[ 8932.737777] Successfully completed within time limit, User:devika
```

Test Case 02: Invalid kernel version (Failed)

Setting kernel version =5.8 instead of compiled 6.11

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo insmod mymodule.ko kernel_version=5,8 time_limit=60
insmod: ERROR: could not insert module mymodule.ko: Invalid parameters
```

Test Case 03: Implementing write before read (Failed)

1. Insert module

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo insmod mymodule.ko kernel_version=6,11 time_limit=60
```

2. Create device

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo mknod /dev/mychardev c 235 0
```

3. Perform write

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ echo "devika" | sudo tee /dev/mychardev
devika
tee: /dev/mychardev: Invalid argument
```

4. Remove module

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo rmmod mymodule
```

5. Check final output

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo dmesg | tail
[11197.607697] Device registered, Major=235, Minor=0, Time=60
[11206.102643] Write attempted before read
[11223.610074] Failure: all conditions not satisfied
```

Conclusion: logs show failure since write was attempted before read operation

Test Case 04: Time limit exceeded (Failed)

1. Insert module

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo insmod mymodule.ko kernel_version=6,11 time_limit=10
```

2. Check logs for device registration

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo dmesg | tail
[11541.435738] Device registered, Major=235, Minor=0, Time=10
```

3. Create device

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo mknod /dev/mychardev c 235 0
```

4. Perform read

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ cat /dev/mychardev
```

5. Wait (to exceed time limit)

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sleep 15
```

6. Perform write

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ echo "devika" | sudo tee /dev/mychardev
devika
```

7. Remove module

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo rmmod mymodule
```

8. Check final output

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo dmesg
[11541.435738] Device registered, Major=235, Minor=0, Time=10
[11558.972890] Device read completed successfully
[11593.549232] Username recieved: devika
[11598.115936] Failure: all conditions not satisfied
```

Test Case 05: Only read performed (Failed)

1. Insert module

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo insmod mymodule.ko kernel_version=6,11 time_limit=60
```

2. Check logs for device registration

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo dmesg | tail
[12218.837202] Device registered, Major=235, Minor=0, Time=60
```

3. Create device

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo mknod /dev/mychardev c 235 0
```

4. Perform read

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ cat /dev/mychardev
```

5. Remove module (skip write)

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo rmmod mymodule
```

6. Check logs

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo dmesg | tail
[12218.837202] Device registered, Major=235, Minor=0, Time=60
[12229.794097] Device read completed successfully
[12236.092719] Failure: all conditions not satisfied
```

Test Case 06: Only write performed (Failed)

1. Insert module

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo insmod mymodule.ko kernel_version=6,11 time_limit=60
```

2. Check logs for device registration

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo dmesg | tail
[12687.943864] Device registered, Major=235, Minor=0, Time=60
```

3. Create device

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo mknod /dev/mychardev c 235 0
```

4. Perform write (skip read)

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ echo "devika" | sudo tee /dev/mychardev
devika
tee: /dev/mychardev: Invalid argument
```

5. Remove module

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo rmmod mymodule
```

6. Check logs

```
kdevika@fedora:~/Documents/OS_S4/Assignment_1$ sudo dmesg
[12687.943864] Device registered, Major=235, Minor=0, Time=60
[12699.456262] Write attempted before read
[12705.804620] Failure: all conditions not satisfied
```

Conclusion

The character device driver was successfully implemented and tested for both valid and invalid scenarios. The results confirm correct behavior in terms of kernel version validation, operation order enforcement, and time-based constraints.